Time limit: 1.00 s **Memory limit:** 512 MB

You and some monsters are in a labyrinth. When taking a step to some direction in the labyrinth, each monster may simultaneously take one as well. Your goal is to reach one of the boundary squares without ever sharing a square with a monster.

Your task is to find out if your goal is possible, and if it is, print a path that you can follow. Your plan has to work in any situation; even if the monsters know your path beforehand.

Input

The first input line has two integers n and m: the height and width of the map.

After this there are n lines of m characters describing the map. Each character is `.` (floor), `#` (wall), `A` (start), or `M` (monster). There is exactly one `A` in the input.

Output

First print "YES" if your goal is possible, and "NO" otherwise.

If your goal is possible, also print an example of a valid path (the length of the path and its description using characters `D`, `U`, `L`, and `R`). You can print any path, as long as its length is at most $n \cdot m$ steps.

Constraints

- $1 \le n, m \le 1000$

Example

Input:
```
5 8
########
#M..A..#
#.#.M#.#
#M#..#..
#.######
```

Output:
```
YES
5
RRDDR
```


Graph Algorithms

...
Message Route
Building Teams
Round Trip
Monsters
Shortest Routes I
Shortest Routes II
High Score
Flight Discount
...

Your submissions